Exercise Session 9: Magnetostatics 2

Question 1:

A distribution of charge may lie along a wire, on a surface, or throughout a volume.

a) Explain the difference between charge density compared to line current density in 1D, and surface and volume current densities in 2D and 3D, respectively.

b) For each case below, state the appropriate quantity used to describe mobile charge and explain what it means physically:

- a wire
- a thin conducting sheet
- a bulk conducting material

c) Write the relation between the moving charge density and velocity for each case.

d) Write the expression for the magnetic force on each type of current distribution.

e) Compare these mobile charge quantities with the expressions used for constant charge flow in 1D, 2D, and 3D.

f) In one or two sentences, explain why the same symbols λ, σ, and ρ can appear in both electrostatics and magnetostatics, but do not represent exactly the same physical situation.

Question 2:

A volume V is enclosed by a closed surface S. Charge may flow into or out of the volume.

a) If more charge flows out through the surface than flows in, what must happen to the amount of charge inside the volume?

b) Use this idea to derive the continuity equation involving the divergence theorem,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

c) Explain in words what each term means.

d) In magnetostatics, why does this reduce to

$$\nabla \cdot \mathbf{J} = 0?$$

e) Give a physical example of what would happen if $\nabla \cdot \mathbf{J} \neq 0$ in some region.

Question 3:

A long straight conductor carries a steady current in the $+\hat{z}$ direction. Its cross section is a rectangle of width a and height b, as illustrated in Figure 1.

a) If the current is uniformly distributed over the cross section, find the volume current density $\mathbf{J}$.
b) Suppose instead that the current density varies with x according to

$$\mathbf{J}(x) = kx\hat{z}, 0 \leq x \leq a.$$

Find the total current I.

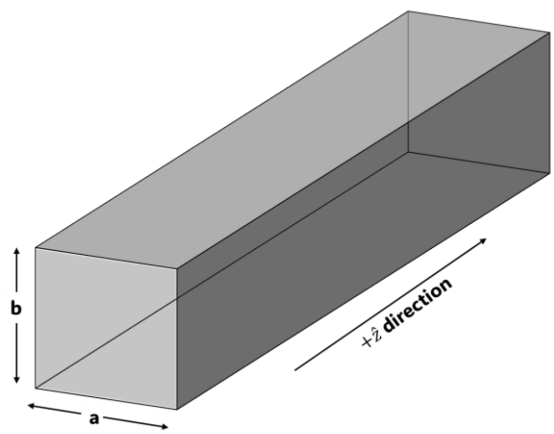

Figure 1. A long straight conductor that carries a steady current in the $+\hat{z}$ direction.

Question 4:

A square loop of wire carries a steady current I. Let R be the perpendicular distance from the center of the loop to any side. Find the magnetic field at the center of the square.

Hints: The field from each side points in the same perpendicular direction at the center, so the four contributions add directly. And treat one side as a finite straight wire and use the result from Example 5.5:

$$B = \frac{\mu_0 I}{4\pi s}(\sin\,\theta_2 - \sin\,\theta_1).$$

Question 5:

Two infinitely long parallel conductors are separated by a distance d where the left wire carries current $2I$ upward and the right wire carries current $3I$downward.

Let, P be a point a distance a to the left of the wire carrying $2I$, Q be a point a distance a to the left of the wire carrying $3I$, with $0 < a < d$ and M be the midpoint between the two wires.

Consider the unit vector describing the direction towards out of the page as $+\hat{k}$.

Find:

a) the net magnetic field at P,

b) the net magnetic field at Q,

c) the net magnetic field at M,

d) the force per unit length between the two wires, and state whether it is attractive or repulsive.